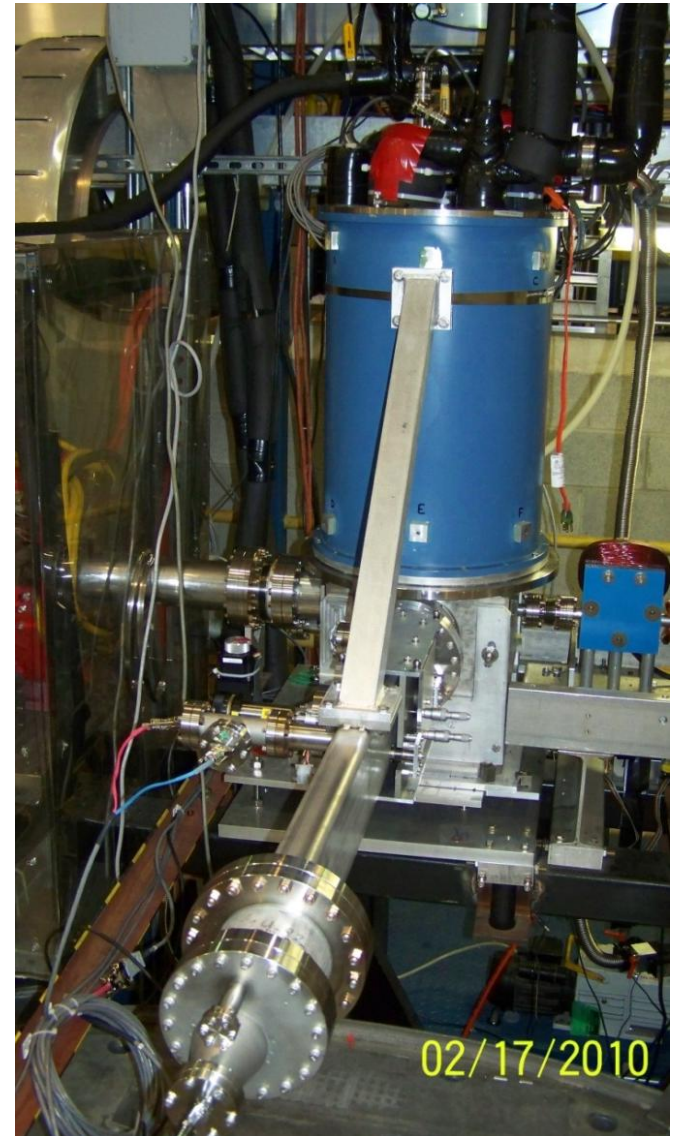


Møller Polarimetry for PREX (status report)

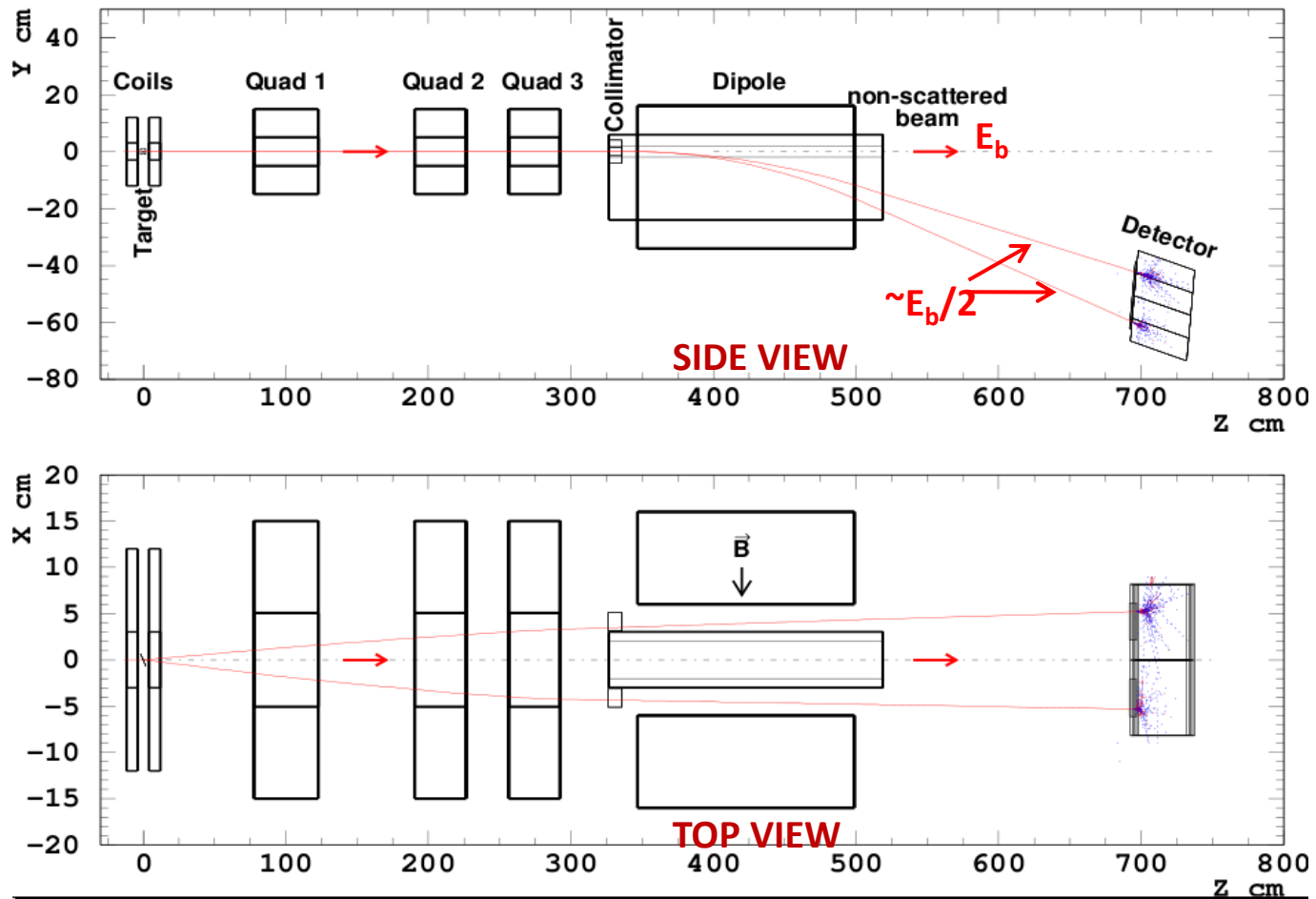
O. Glamazdin
Kharkov Institute of Physics and Technology
Kharkov, Ukraine

What was done?

1. Polarized electron target "brute force" 3T (smaller thickness, lower heating)
2. Segmented aperture detector (higher rate)
3. New fast DAQ based on FADC (higher rate, smaller dead time, more information etc.)



Møller Polarimeter Layout



$$P_{\text{beam}} = A_{\text{sym}} / A_{\text{nPow}} \times P_{\text{targ}}$$

Møller Problems after PREX

1. Magnet alignment
(→ Acceptance → Analyzing Power)
2. Quads discrepancy
(→ Acceptance → Analyzing Power)
3. Targets discrepancy
(→ Target polarization)
4. Targets saturation
(→ Target polarization)
5. Old DAQ vs. FADC

Møller Magnet Alignment

PREX

March 30, 2010:

$dX = -0.3\text{mm}$ $dY = -1.1\text{mm}$ $d\text{Horiz} = -1.8\text{mrad}$ $d\text{Vert} = +4.4\text{mrad}$

April 17, 2010:

$dX = -0.2\text{mm}$ $dY = -1.7\text{mm}$ $d\text{Horiz} = -1.6\text{mrad}$ $d\text{Vert} = +6.1\text{mrad}$

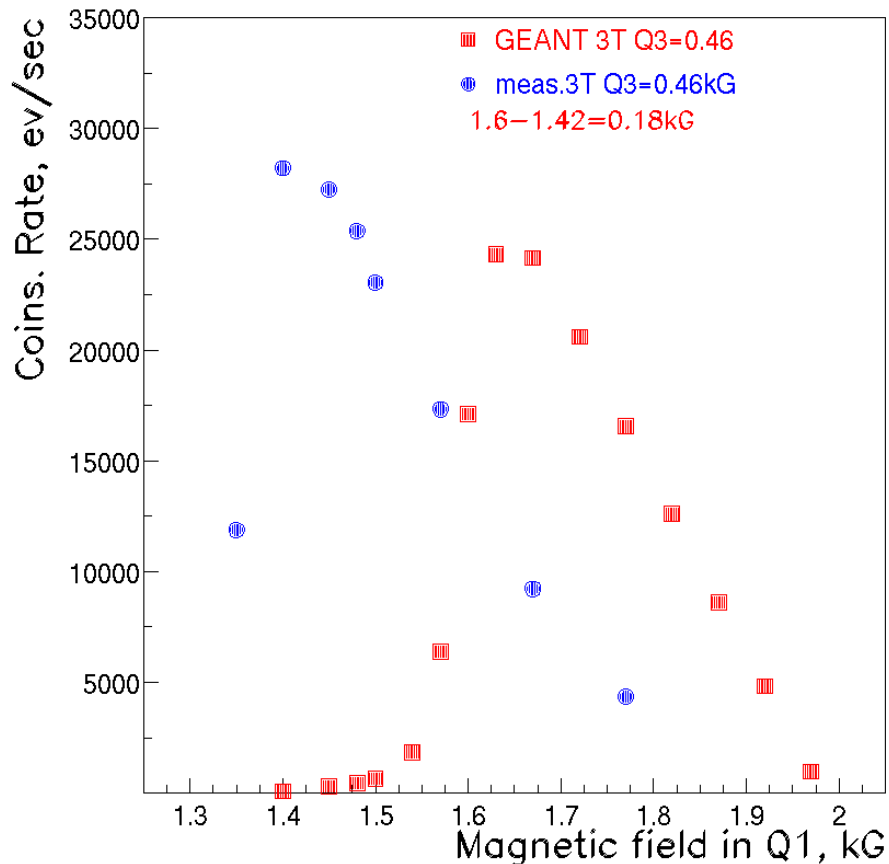
DVCS

September 30, 2010:

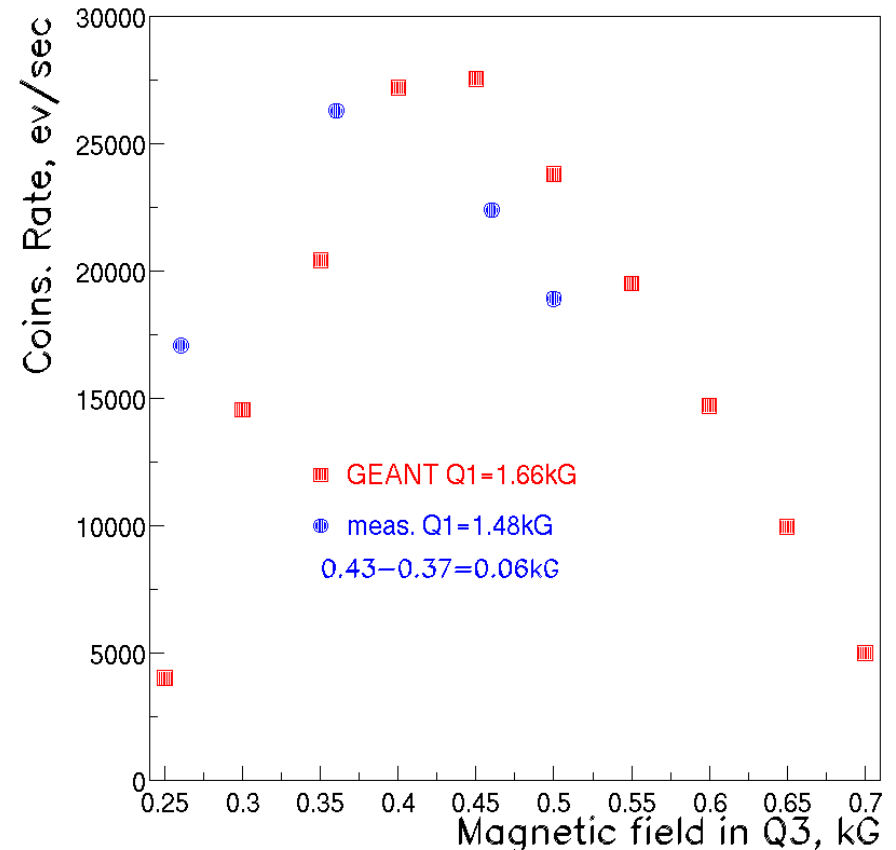
$dX = -0.2\text{mm}$ $dY = -1.0\text{mm}$ $d\text{Horiz} = -0.2\text{mrad}$ $d\text{Vert} = +2.9\text{mrad}$

Quads Scan PREX

Quad Q1 Scan. Measurement vs. GEANT

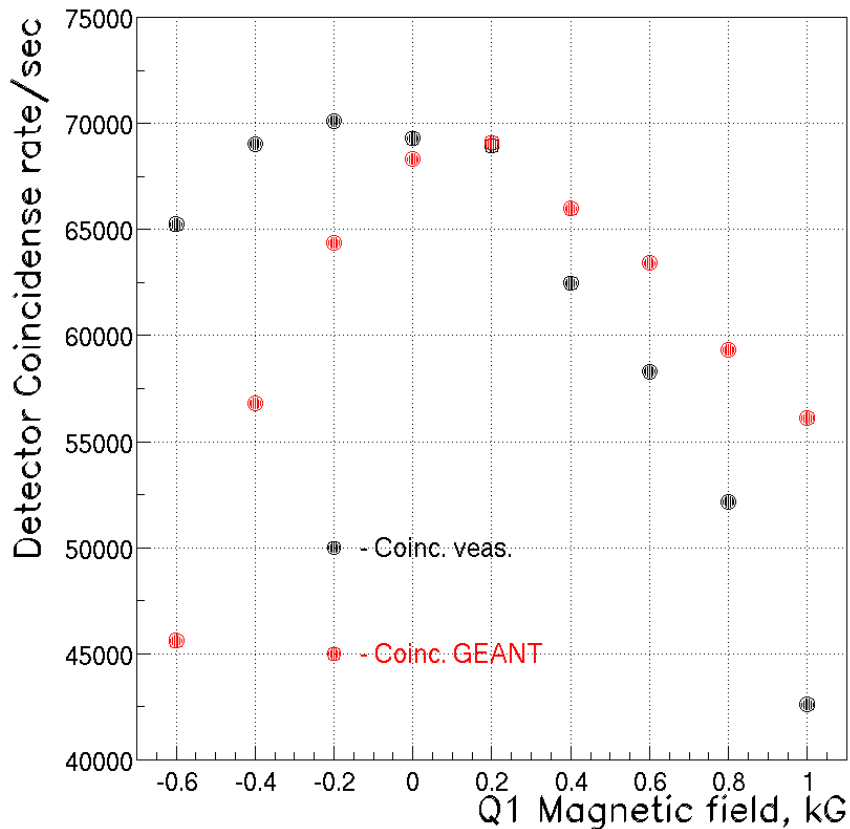


Quad Q3 Scan. Measurement vs. GEANT

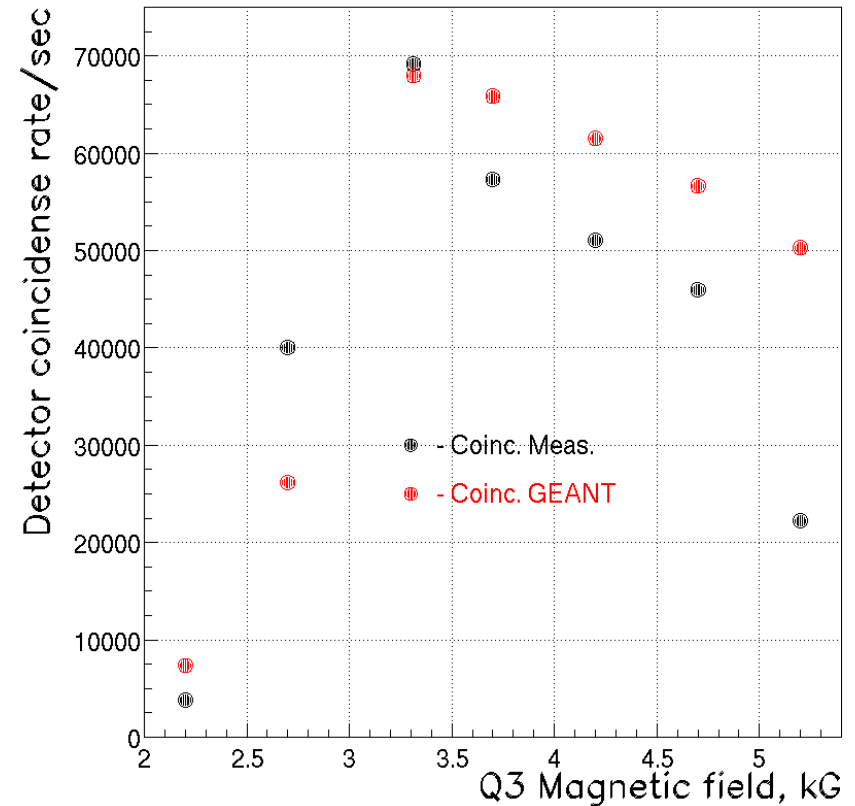


Quads Scan DVCS

DVCS 1st Moller Quad Scan



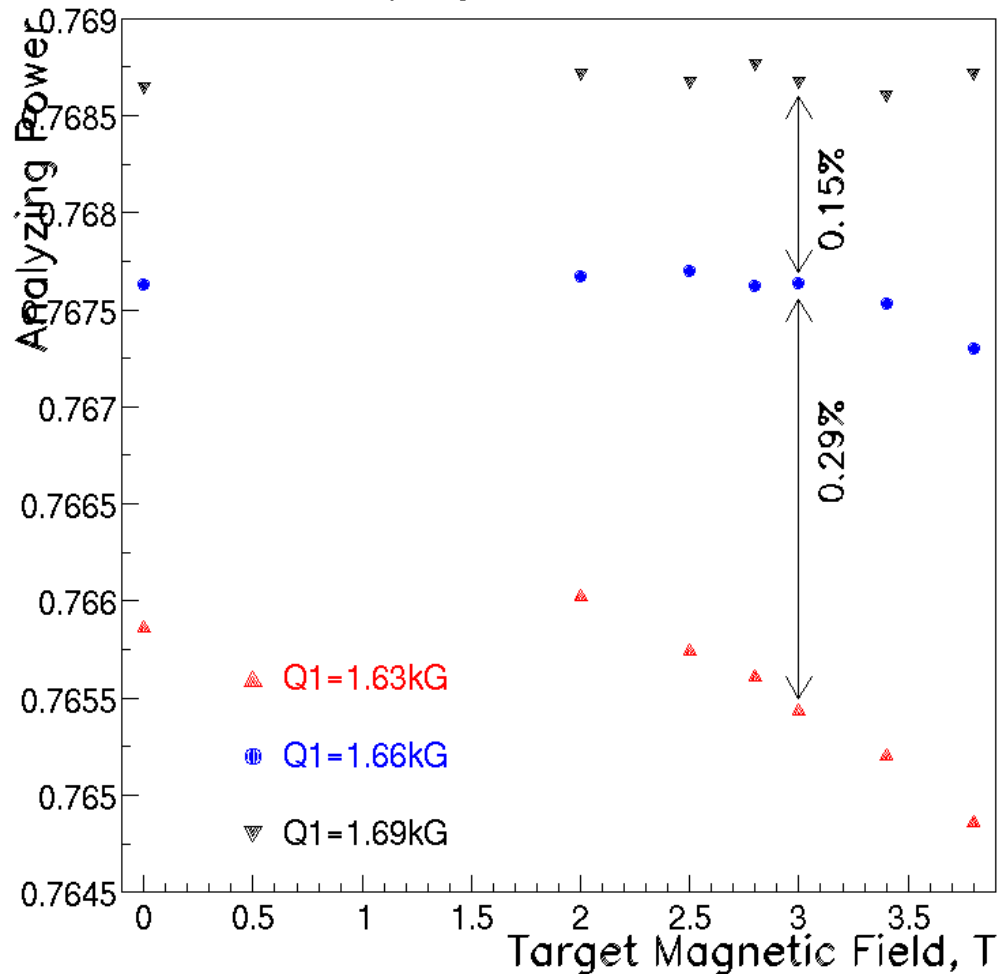
DVCS 3rd Moller Quad Scan



- Request quads power supply test (current output vs. current read out)
- New quads scan (3.5GeV, March 2011)

Analyzing Power

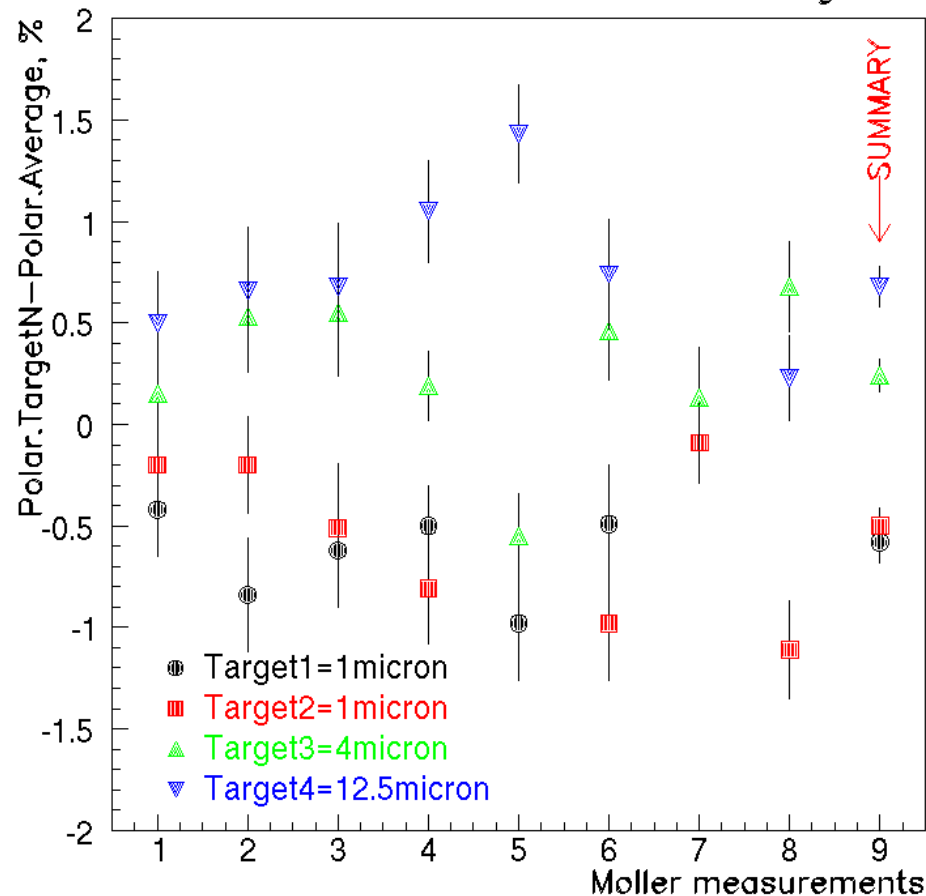
PREX Analyzing Power vs. Saturated Field



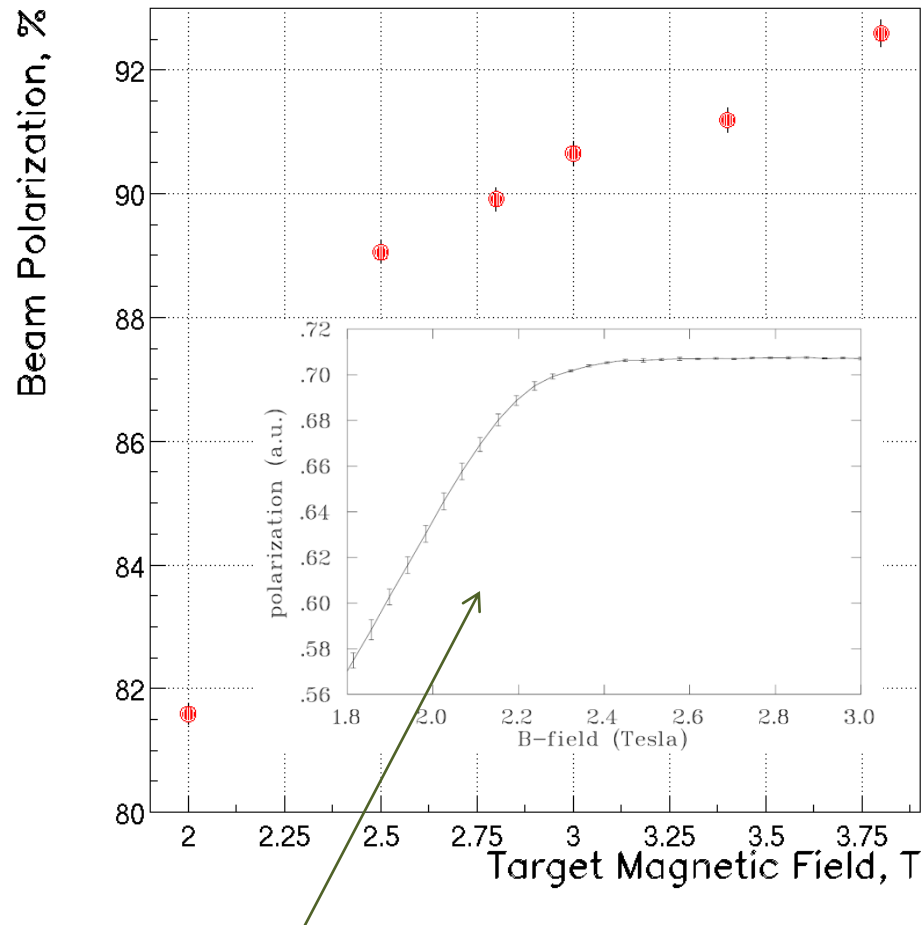
Systematic error $\pm 0.3\%$

Target Polarization

Beam Polarization vs. Moller Target



PREX Target Saturation Test

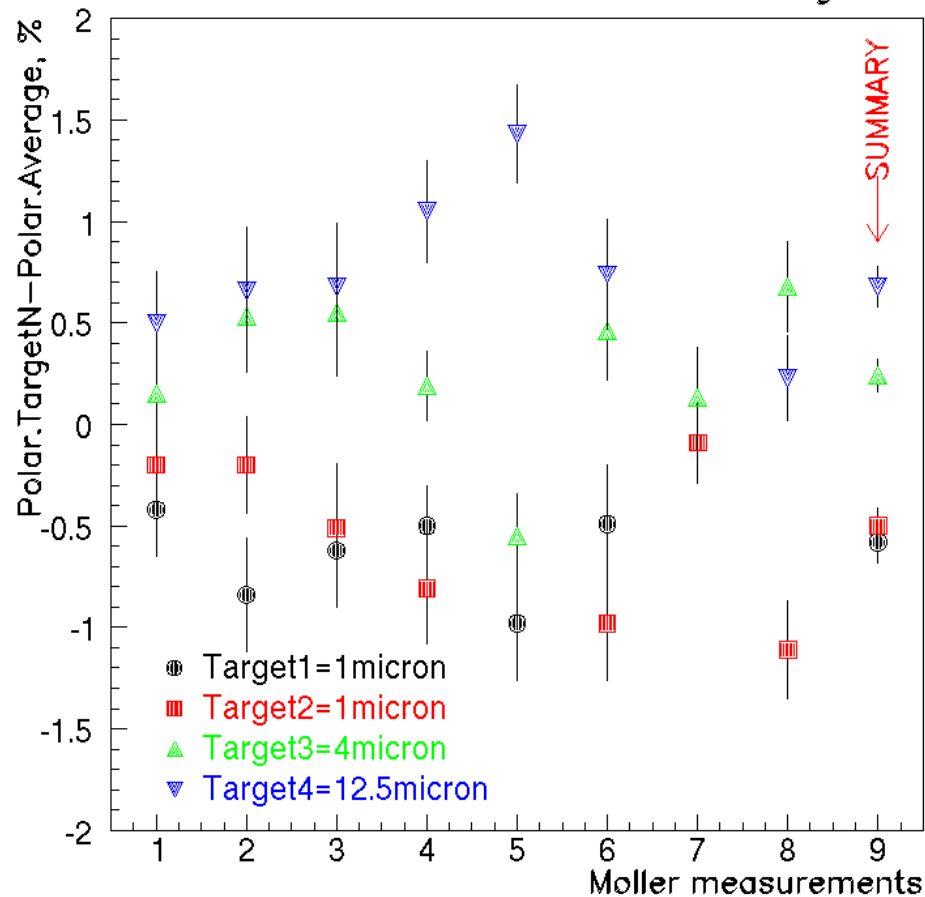


M. Loppacher "Møller Polarimetry for CEBAF Hall C" PhD thesis, Basel 1996

Target Polarization

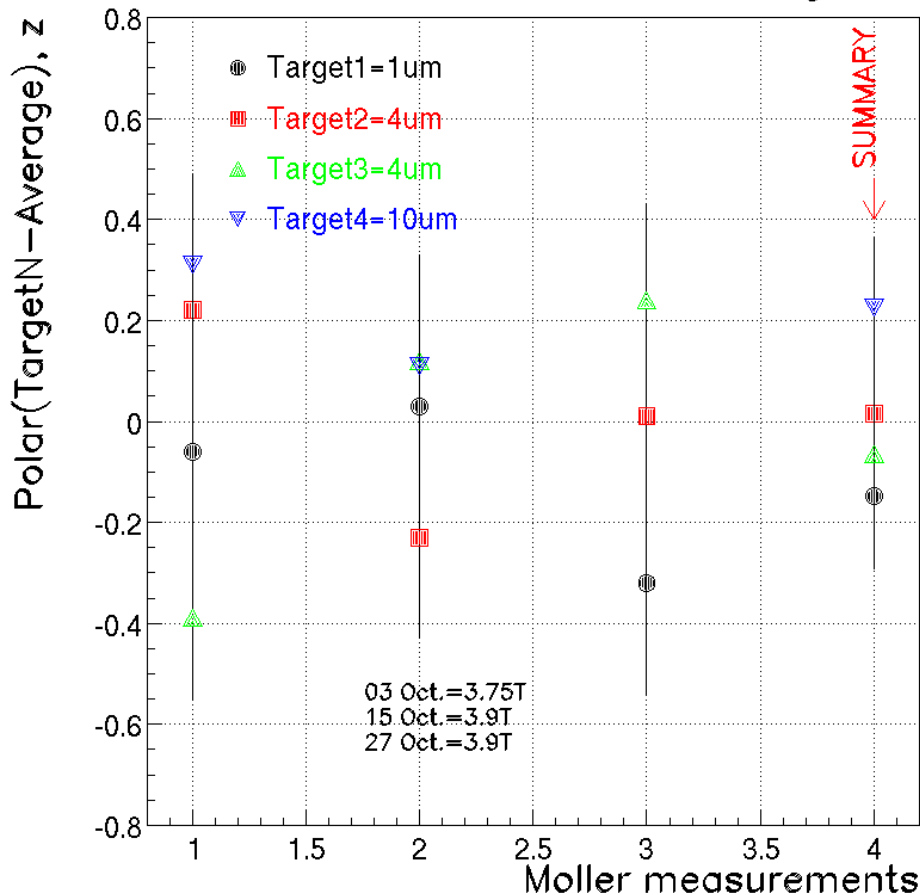
PREX (3T)

Beam Polarization vs. Moller Target



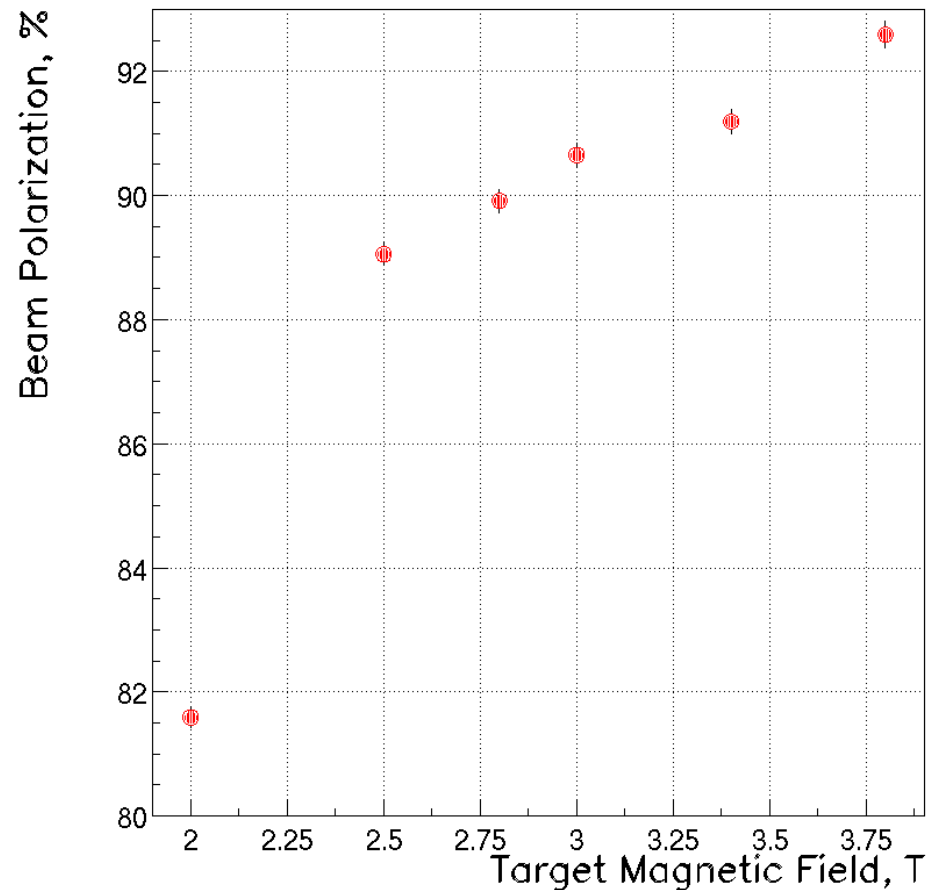
DVCS (3.75-3.9T)

DVCS Beam Polarization vs. Moller Target

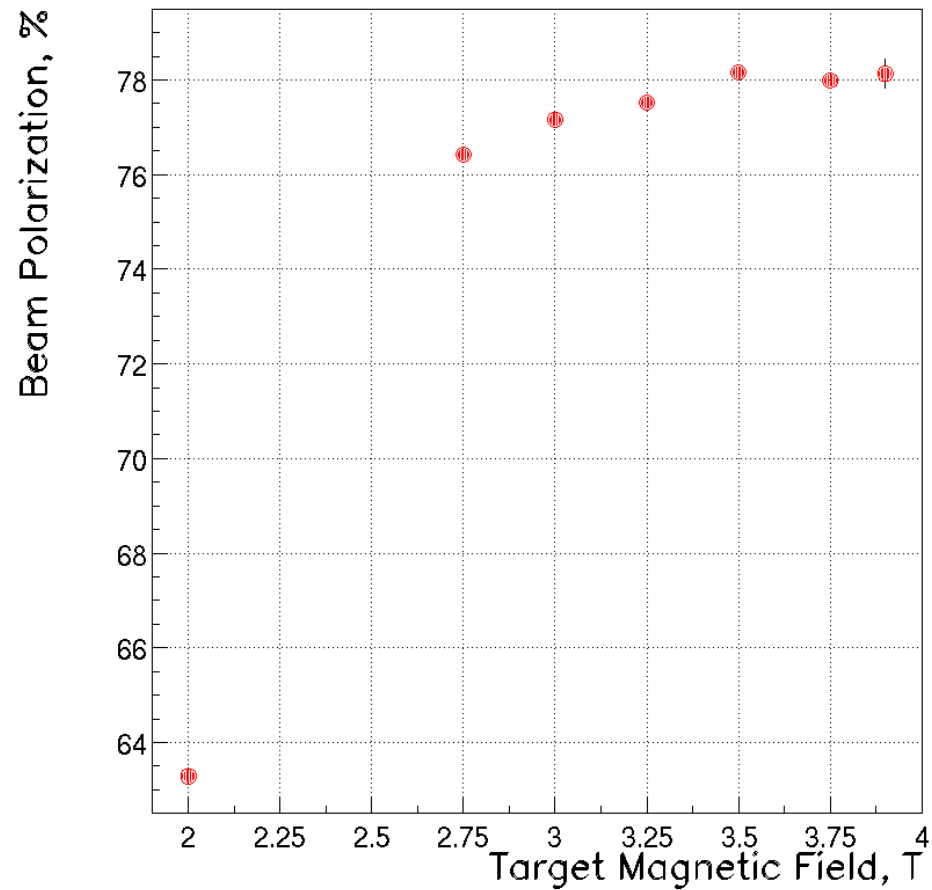


Target Saturation

PREX Target Saturation Test

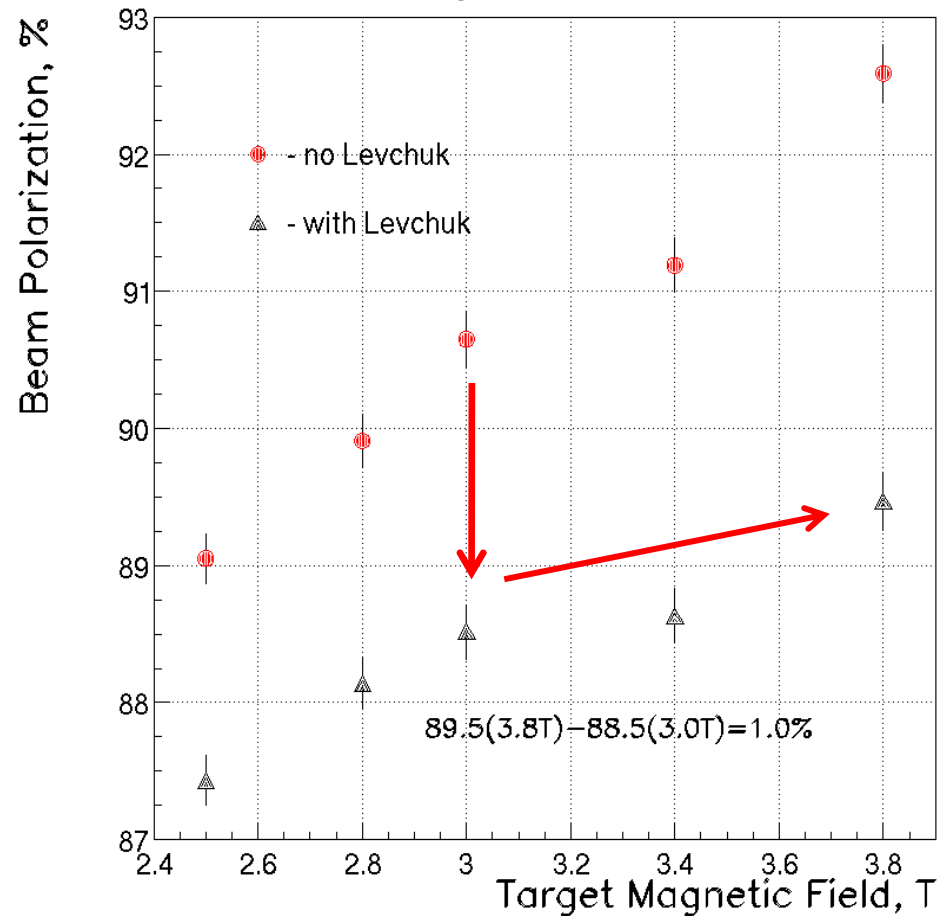


DVCS Target Saturation Test

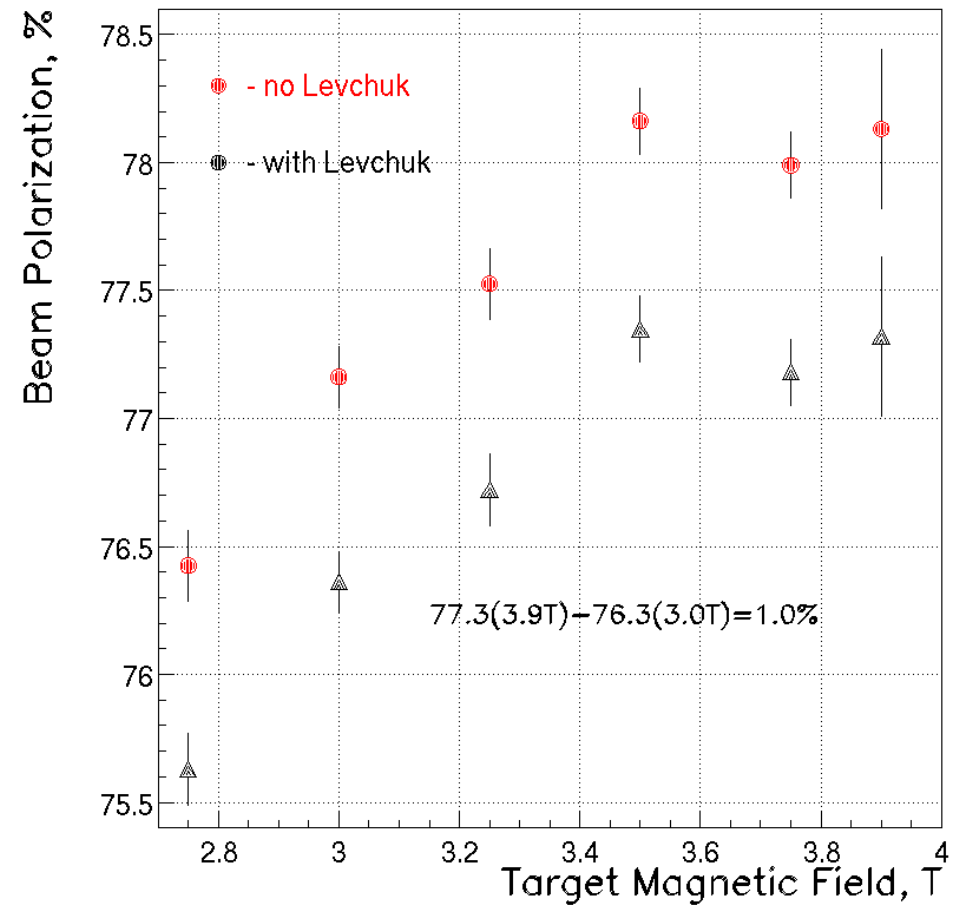


Levchuk-effect

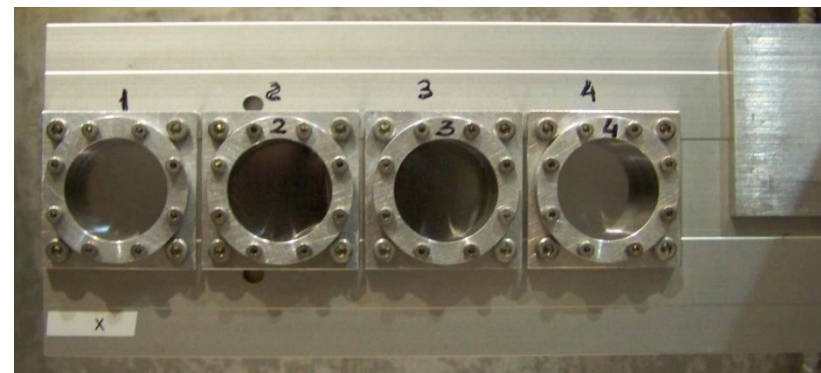
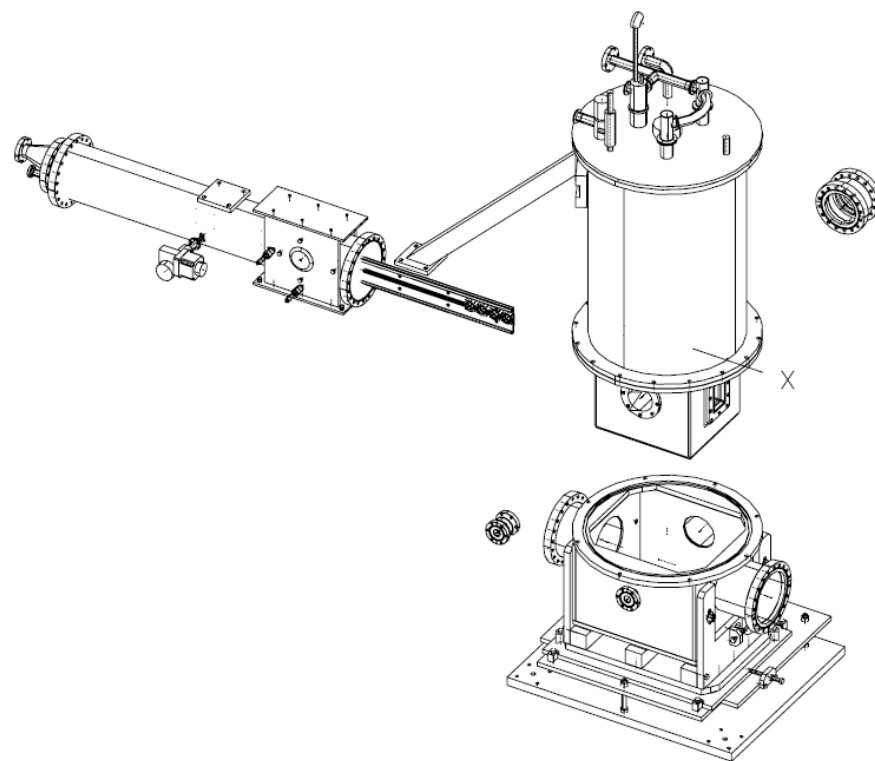
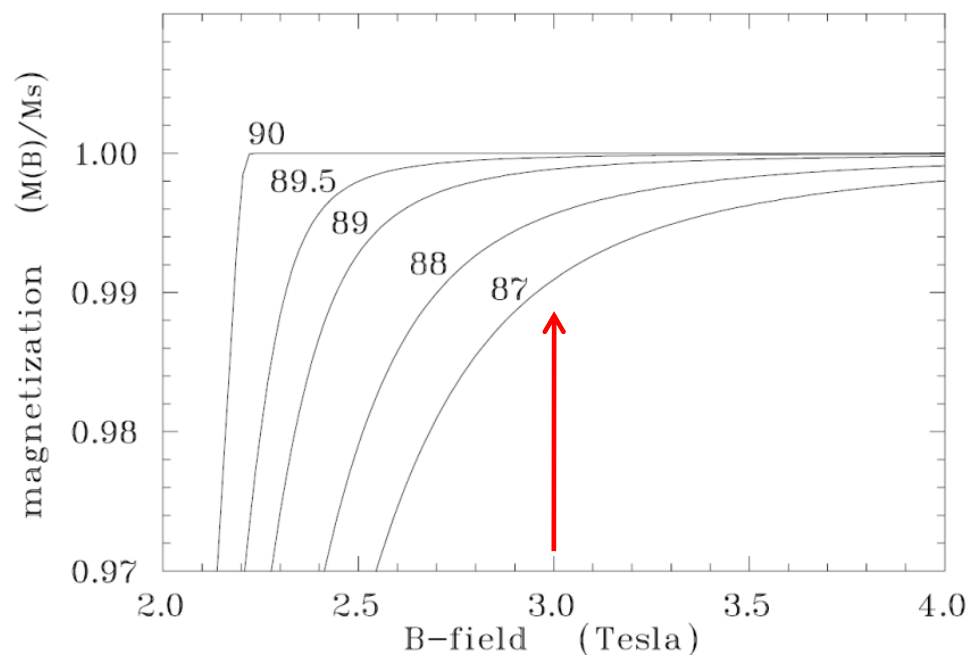
PREX Target Saturation Test



DVCS Target Saturation Test



Target Warping+Misalignment



M. Loppacher "Møller
Polarimetry for CEBAF Hall C"
PhD thesis, Basel 1996

Systematic Error for PREX

Variable	PREX(old DAQ)
<i>Iron Foil Polarization</i>	0.25%
<i>Targets Discrepancy</i>	0.5%
<i>Target Saturation</i>	0.3%
<i>Analyzing power</i>	0.3%
<i>Levchuk effect</i>	0.5%
<i>Target temperature</i>	0.02%
<i>Dead time</i>	0.3%
<i>Background</i>	0.3%
<i>Others</i>	0.5%
<i>Total</i>	1.1%

High Current Measurements

1. EXPERT mode (pulse beam)

- 6-9 μA beam current ;
- special dead time study is required.

2. Bunch suppression (499MHz)

- have to turn injector pre-buncher OFF - INVASIVE;
- specially dedicated measurement;
- special study of cavities load;
- bunch of cross-calibrations;
- can get more new questions than answers

"Brute Force" target

- maximal CW current for Møller is 3 μA
- fully open slit
- Hall A energy lock available

Møller with solid target = low current method

SUMMARY

- 1. We understand Møller analyzing power with reasonable accuracy*
- 2. We understand target polarization with reasonable accuracy*
- 3. We have to re-analyze all Møller data for PREX with Levchuk and saturation corrections*
- 4. Møller systematic error for PREX is 1.1%*